

Omar Khaled

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SUMMARY

Data Scientist and Artificial Intelligence Engineer focused on developing applied AI and data analytics solutions across machine learning, deep learning, computer vision, NLP, predictive analytics, and data visualization. Experienced in collecting, cleaning, interpreting, and modeling data using Python, SQL, TensorFlow, OpenCV, Spark MLlib, and statistical analysis workflows. Published AI research in IEEE and MDPI venues, with projects involving algorithms, business insights, process automation, and stakeholder-ready dashboards.

TECHNICAL SKILLS

- **Programming Languages:** Python, SQL, Java, JavaScript, C++
- **AI & Machine Learning:** Machine Learning, Deep Learning, RAG, LLMs, Predictive Analytics, Algorithms, Model Development
- **Statistics & Mathematics:** Statistical Analysis, Mathematical Modeling, Linear Algebra, Calculus, Classification, Regression
- **Data Science & Visualization:** Data Cleaning, Data Preprocessing, Feature Engineering, Data Interpretation, Data Visualization, Business Intelligence
- **Computer Vision & NLP:** Computer Vision, Image Classification, Object Detection, NLP, Sentiment Analysis, Text Classification
- **Libraries & Tools:** TensorFlow, OpenCV, Scikit-learn, Pandas, NumPy, Spark MLlib, FAISS, Plotly Dash, Streamlit, Gradio, Jupyter
- **Databases & Backend:** MySQL, SQLite, MongoDB, REST APIs, Git, GitHub

EDUCATION

Arab Academy for Science, Technology and Maritime Transport

Alamein, Egypt

Bachelor of Science in Artificial Intelligence, Data Science Specialization

(2023 – 2026)

GPA: 3.93/4.00 | Excellent with Honors | Ranked 1st in Data Science Department

Graduation Project Grade: Excellent

Graduated: Aug 2026

PROJECTS

[OmniRAG Assistant](#) | Python, Streamlit, Ollama, FAISS, LLaVA, RAG

(Feb 2026)

- Built a multimodal RAG assistant supporting 6 input types: PDF, CSV, arXiv, image, audio upload, and live microphone.
- Implemented 2 chat modes and a 5-step retrieval pipeline using FAISS, Sentence-Transformers, and local LLMs.
- Added summarization, text-to-speech, audio transcription, and PDF chat export for non-technical users.

[ReviewPulse: Amazon Sentiment Analytics](#) | Spark MLlib, Kafka, MongoDB

(Apr 2025)

- Developed a real-time sentiment analytics pipeline for large-scale Amazon reviews using Spark MLlib and Kafka.
- Processed 2,000,000 Amazon review rows and trained sentiment models on 816,926 balanced samples.
- Compared 4 ML models and achieved 81.88% accuracy and 0.8187 F1 using optimized Logistic Regression.
- Visualized real-time customer feedback insights through a Plotly Dash dashboard for stakeholder reporting.

[Driver Monitoring System](#) | TensorFlow, Computer Vision, Gradio

(Jun 2025)

- Built a deep learning system to classify distracted driving behavior across 10 driver behavior classes.
- Compared 4 deep learning models with ensemble inference, confidence scoring, risk assessment, and prediction logging.

PUBLICATIONS

- O. Khaled et al., "EASE-PVNet: Robust Periocular Identity Verification Across Pre- and Post-Operative Facial Images," Machine Learning and Knowledge Extraction, 2026.
DOI: [10.3390/make8060169](#)
- O. Khaled et al., "Deep CNN and SimpleRNN Architectures for Bi-Handed Arabic Alphabet Hand-Gesture Recognition," ICEENG, 2026.
DOI: [10.5281/zenodo.20161073](#)
- O. Khaled et al., "An Intelligent Web-Based Multi-Agent Framework for Automated University Timetabling via Constraint-Aware Optimization," ICMISI, IEEE, 2026.
DOI: [10.5281/zenodo.20159940](#)

EXPERIENCE

- Application Development and Integration Intern | [EJADA](#)**

(Aug 2025 – Sep 2025)

 - Developed Java backend components for process integration using REST APIs, microservices, and database workflows
 - Implemented API management, testing, and integration tasks for a final development project.
 - Debugged backend workflows and improved application development structure.
- Machine Learning & Frontend Intern | [CodeAlpha](#)**

(Aug 2024 – Sep 2024)

 - Completed a 2-track internship covering machine learning, data preprocessing, model evaluation, and frontend data visualization
 - Built ML projects using Python, data preprocessing, model training, and evaluation.
 - Analyzed real-world datasets to improve model development and prediction workflows.
 - Developed responsive web interfaces using HTML, CSS, and JavaScript.

CERTIFICATIONS

- [Databases and SQL for Data Science with Python](#) | IBM, Coursera

(Jul 2024)
- [AI Essentials](#) | Google, Coursera

(Jul 2024)
- [Prompt Engineering](#) | Dubai Future Foundation and Dubai Centre for AI

(May 2025)
- [Data, Data, Everywhere](#) | Google, Coursera

(Jul 2024)

LEADERSHIP & VOLUNTEERING

- Team Leader, AAST RALLY Competition | AASTMT Society**

(Apr 2023 – Sep 2025)

 - Led team planning, coordination, and project progress follow-up.
- Team Volunteer, ECPC Egyptian Collegiate Programming Contest**

(Jul 2024)

 - Supported participant guidance, logistics, and competition execution.
- Technical Event Volunteer, RoboCup Junior Egypt**

(Apr 2025)

 - Supported technical activities, operations, and participant guidance.
- HR Member, Entrepreneurship Society Alamein**

(Mar 2023 – Jan 2026)

 - Supported internal communication, member coordination, follow-up, and team engagement across 2+ years